

NATIONAL UNIVERSITY OF SINGAPORE

NATIONAL UNIVERSITY OF SINGAPORE READING ASSESSMENT 1 FOR Semester 1 AY2024/25

CS1101S Programming Methodology

2024

Time Allowed: 45 minutes

INSTRUCTIONS TO CANDIDATES

1. This question paper contains 20 questions and comprises 11 printed pages, including this page.
2. You are also provided with an answer sheet to write your answers. It comprises 2 printed pages.
3. Use a pen or pencil to write your Student Number in the designated space on the front page of the answer sheet, and shade the corresponding circle completely in the grid for each digit or letter. **DO NOT WRITE YOUR NAME.**
4. You must submit only the answer sheet and no other documents.
5. All questions must be answered in the space provided in the answer sheet; no extra sheets will be accepted as answers.
6. For multiple choice questions, shade in the circle of the correct answer completely.
7. Each question has one correct answer. One mark is awarded for each correct answer and there is no penalty for a wrong answer.
8. The full score of this assessment is **19 marks**.
9. Answer all questions.
10. This is a **closed-book** assessment.

1. What is the alphabet letter that is in the box at the top-right corner on the front page of this question paper? Important: Please make sure your answer is correct because it determines how your answers to all the subsequent questions will be graded.

- (a.) A
- (b.) B
- (c.) C
- (d.) D

Section A

In the following questions, the pre-declared `print` function is used in the Python programs. The `print` function prints the value of its input argument to the console. For example, `print(2 * 5)` prints 10.

2. What is the value printed when the following Python \$1 program is evaluated?

```
x = 1

def f(x):
    return x + 1
print(f(2))
```

- (a.) Error: name x is redeclared
- (b.) 1
- (c.) 3
- (d.) Error: name f is used before declared
- (e.) 2
- (f.) Error: name x is used before declared

3. What is the value printed when the following Python \$1 program is evaluated?

```
def f(x):  
    def g(x):  
        return 2 * x  
    return g(2 * x)  
print(f(2))
```

- (a.) Error: name g is redeclared
 - (b.) 2
 - (c.) Error: name x is redeclared
 - (d.) 8
 - (e.) 4
 - (f.) 6
4. What is the value printed when the following Python \$1 program is evaluated?

```
x = 1  
  
def f(z):  
    f = lambda x: x + 1  
    g = lambda y: x  
    return g(f(z))  
print(f(3))
```

- (a.) 1
- (b.) Error: name x is used before declared
- (c.) 4
- (d.) Error: name y is used before declared
- (e.) 3
- (f.) Error: name z is used before declared

5. What is the value printed when the following Python \$1 program is evaluated?

```
x = 1

def f(y):
    x = x
    return x + y
print(f(3))
```

- (a.) Error: name x is redeclared
 - (b.) 4
 - (c.) 3
 - (d.) Error: name x is used before declared, on an earlier line
 - (e.) 1
 - (f.) Error: name x is used before initialized, on the same line
6. What is the value printed when the following Python \$1 program is evaluated?

```
f = lambda g: g(f)
print(f(lambda x: f)(lambda y: y)(lambda z: 1))
```

- (a.) A function
- (b.) A number
- (c.) No result: the evaluation does not terminate
- (d.) Error: name f is used before declared
- (e.) Error: name g is used before declared
- (f.) Error: one or more names is/are redeclared

7. What is the value printed when the following Python §1 program is evaluated?

```
f = lambda x: y
y = 1
g = lambda y: f(y)
print(g(2))
```

- (a.) No result: the evaluation does not terminate
- (b.) 2
- (c.) A function
- (d.) Error: name y is used before declared
- (e.) 1
- (f.) Error: one or more names is/are redeclared

8. What is the value printed when the following Python §1 program is evaluated?

```
def f(n):
    h = lambda x: lambda y: x(x(y))
    return h(lambda x: x + 1)(h(lambda x: 2 * x)(n))
print(f(5))
```

- (a.) 12
- (b.) 28
- (c.) 11
- (d.) None of the other options is the correct answer
- (e.) 23
- (f.) 14
- (g.) 22

Section B

We know that in Python §1 the left operand of a binary operation is evaluated before the right operand, and when there are multiple arguments in a function application, the argument expressions are evaluated from left to right.

In some of the following questions, the pre-declared `print` function is used in the programs. The `print` function prints the value of its input argument to the console.

9. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def fun(n):  
    print(n)  
    return n if n <= 1 else n + fun(n / 2)  
fun(32)
```

- (a.) 3 7 15 31 63
 - (b.) 32 16.0 8.0 4.0 2.0 1.0
 - (c.) 32 16 8 4 2
 - (d.) 1 2 4 8 16 32
 - (e.) 2 4 8 16 32
 - (f.) 63 31 15 7 3
10. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def fun(n):  
    if n <= 1:  
        return n  
    else:  
        result = n + fun(n / 2)  
        print(result)  
        return result  
fun(32)
```

- (a.) 1 3 7 15 31
- (b.) 2 4 8 16 32
- (c.) 63 31 15 7 3
- (d.) 3.0 7.0 15.0 31.0 63.0
- (e.) 31 15 7 3 1
- (f.) 32 16 8 4 2

11. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def fun(x):  
    result = x if x <= 4 else fun(x - 3) + fun(x - 2)  
    print(result)  
    return result  
fun(8)
```

(a.) 2 3 5 3 4 7 12

(b.) 2 3 5 3 4 6 8

(c.) 8 5 2 3 6 3 4

(d.) 12 5 2 3 7 3 4

(e.) 2 3 3 4 5 7 12

(f.) 12 5 7 2 3 3 4

12. What is the sequence of values printed when the following Python \$1 program is evaluated?

```
def D(x):  
    print(x)  
    return x  
  
def fun(x, y):  
    return D(D(4 + D(y)) * D(6 + D(x)))  
fun(D(2), D(3))
```

(a.) 2 3 7 3 8 2 56

(b.) 3 7 2 8 56 2 3

(c.) 2 3 56 7 8 3 2

(d.) 2 3 3 2 7 8 56

(e.) 56 7 3 8 2 2 3

(f.) 2 3 3 7 2 8 56

(g.) 2 3 7 8 3 2 56

(h.) 2 3 56 7 3 8 2

13. What is the sequence of values printed when the following Python 3 program is evaluated?

```
def D(m, x):  
    print(m)  
    return x  
fun = D("1", lambda: D("2", 2) + D("3", 3))  
D("4", 4)  
D("5", D("6", fun()))
```

- (a.) "1" "2" "3" "4" "5" "6"
- (b.) "1" "4" "2" "3" "5" "6"
- (c.) "1" "2" "3" "4" "6" "5"
- (d.) "1" "4" "6" "2" "3" "5"
- (e.) "1" "2" "3" "4" "2" "3" "6" "5"
- (f.) "1" "4" "5" "6" "2" "3"
- (g.) "1" "2" "3" "4" "5" "6" "2" "3"
- (h.) "1" "4" "2" "3" "6" "5"

Section C

14. What kind of process does the following function give rise to for any integer arguments $n > 0$?
Note that `math.floor(x)` returns the greatest integer equal to or less than x .

```
def fun(n):  
    return 0 if n < 1 else (n % 10) + fun(math.floor(n / 10))
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

15. What kind of process does the following function give rise to for any integer arguments $a > 0$ and $b > 0$?

```
def fun(a, b):  
    return 2 * a if b == 0 else fun(a + a, b - 1)
```

- (a.) An iterative process
 - (b.) A process that is neither iterative nor recursive
 - (c.) A substitution process
 - (d.) A recursive process
 - (e.) An infinite process
16. What kind of process does the following function give rise to for any integer argument $n > 3$?

```
def fun(n):  
    return (  
        n  
        if n <= 1  
        else fun(n - 2)  
        if n <= 3  
        else fun(n - 1) + fun(n - 1)  
    )
```

- (a.) An iterative process
 - (b.) A process that is neither iterative nor recursive
 - (c.) A substitution process
 - (d.) A recursive process
 - (e.) An infinite process
17. What kind of process does the function `fun` give rise to for any integer argument $x > 0$?

```
def fun(x):  
    f = lambda x: x + 1  
    return f(x) if f(x) < 2 else f(fun(x - 1))
```

- (a.) An iterative process
- (b.) A process that is neither iterative nor recursive
- (c.) A substitution process
- (d.) A recursive process
- (e.) An infinite process

Section D

18. We specify that the function DD, when applied to any three number arguments a, b, and c, should return the difference between the largest and the smallest values of the three inputs.

```
def DD(a, b, c):  
    return (  
        (a if a > c else c) if a > b else (b if b > c else c)  
    ) - (  
        (a if a < c else c) if a < b else (b if b < c else c)  
    )
```

Which one of the following statements is correct?

- (a.) The function DD does not meet the specification because it causes a run-time error.
 - (b.) The function DD meets the specification.
 - (c.) The function DD does not meet the specification because it has syntax error(s).
 - (d.) The function DD does not meet the specification because it returns incorrect results for all valid argument values.
 - (e.) The function DD does not meet the specification because it returns correct results for some but not all valid argument values.
19. We specify that function NN, when applied to any two boolean values a and b, should return the result of (not a) and b.

```
def NN(a, b):  
    AND = lambda a: lambda b: a and b  
    NOT = lambda a: lambda: not a  
    return AND(NOT(a))(b)
```

Which one of the following statements is correct?

- (a.) The function NN meets the specification.
- (b.) The function NN does not meet the specification because it has syntax error(s).
- (c.) The function NN does not meet the specification because it returns incorrect results for all valid argument values.
- (d.) The function NN does not meet the specification because it returns correct results for some but not all valid argument values.
- (e.) The function NN does not meet the specification because it causes a run-time error.

20. This question has been voided.

We specify that function CC, when applied to any three positive integer arguments a, b, and n, should return the number of integers between, and inclusive of, a and b, that are divisible by n.

```
def CC(a, b, n):  
    return (  
        0  
        if a > b  
        else 1 + CC(a + n, b, n)  
        if a % n == 0  
        else CC(a + n - a % n, b, n)  
    )
```

Which one of the following statements is correct?

- (a.) The function CC meets the specification.
- (b.) The function CC does not meet the specification because it causes a run-time error.
- (c.) The function CC does not meet the specification because it returns correct results only when $a \leq b$.
- (d.) The function CC does not meet the specification because it leads to infinite recursion for some inputs.
- (e.) The function CC does not meet the specification because it returns correct results only when $a > b$.

END OF PAPER